ENERGY INSTITUTE
Colorado State University

Economics of Plug in Vehicles

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Vehicle Economics and Distributed Generation

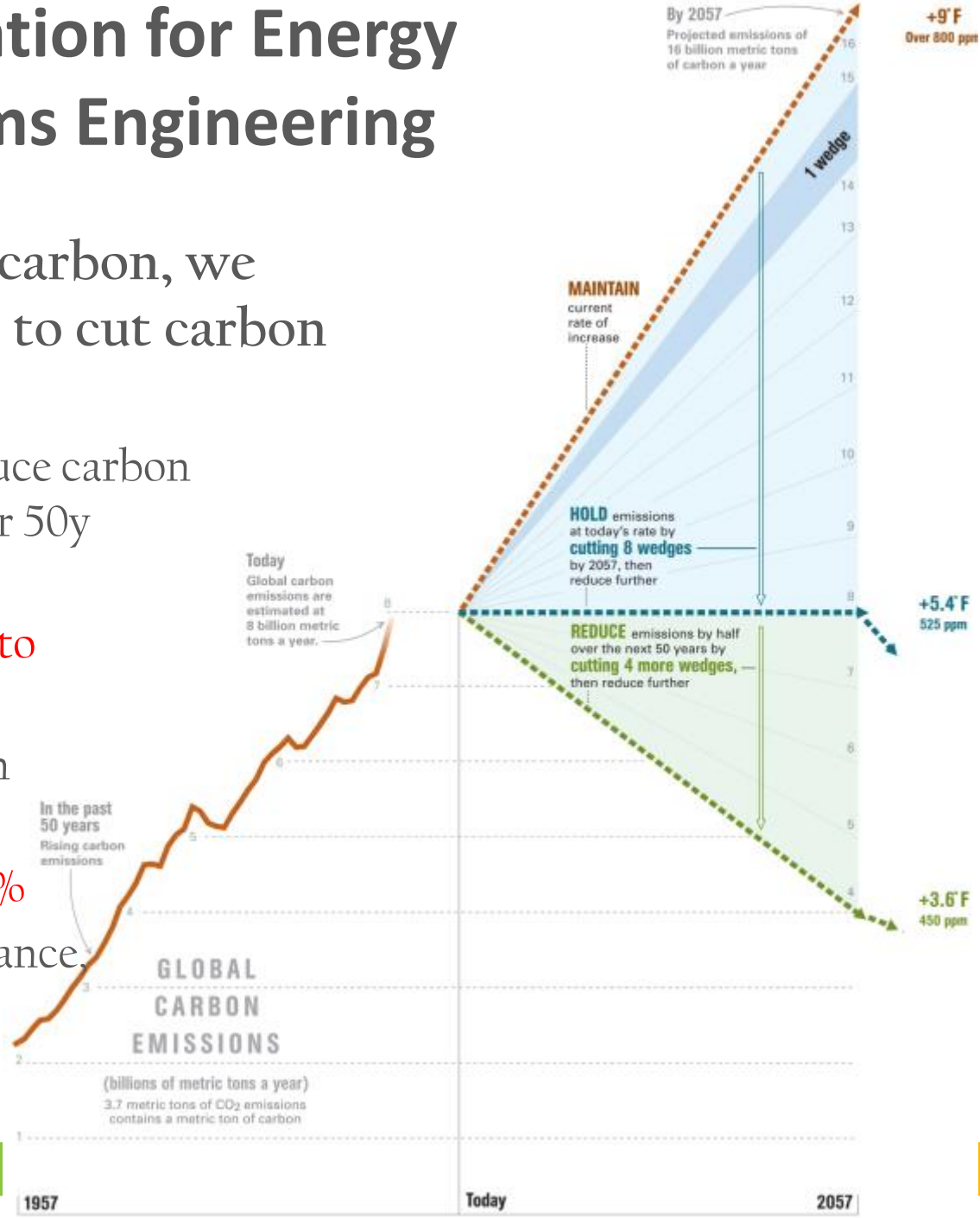
- Contents
  - Background
    - Vehicle Economics
  - Metrics of economic benefit for hybrid vehicles
  - Conclusions

Motivation for Energy Systems Engineering

- To stabilize atmospheric carbon, we must develop 8 strategies to cut carbon emissions

  Each strategy must grow to reduce carbon emissions by 1 GtC per year after 50y of growth

  - Triple vehicle fuel economy to 60+mpg
  - *And* 12x biofuels production
  - *And* improve efficiency of electricity production to 60%
  - *And* increase lighting, appliance, HVAC efficiency by 25%
  - *And* 12 more to hold today's baseline …

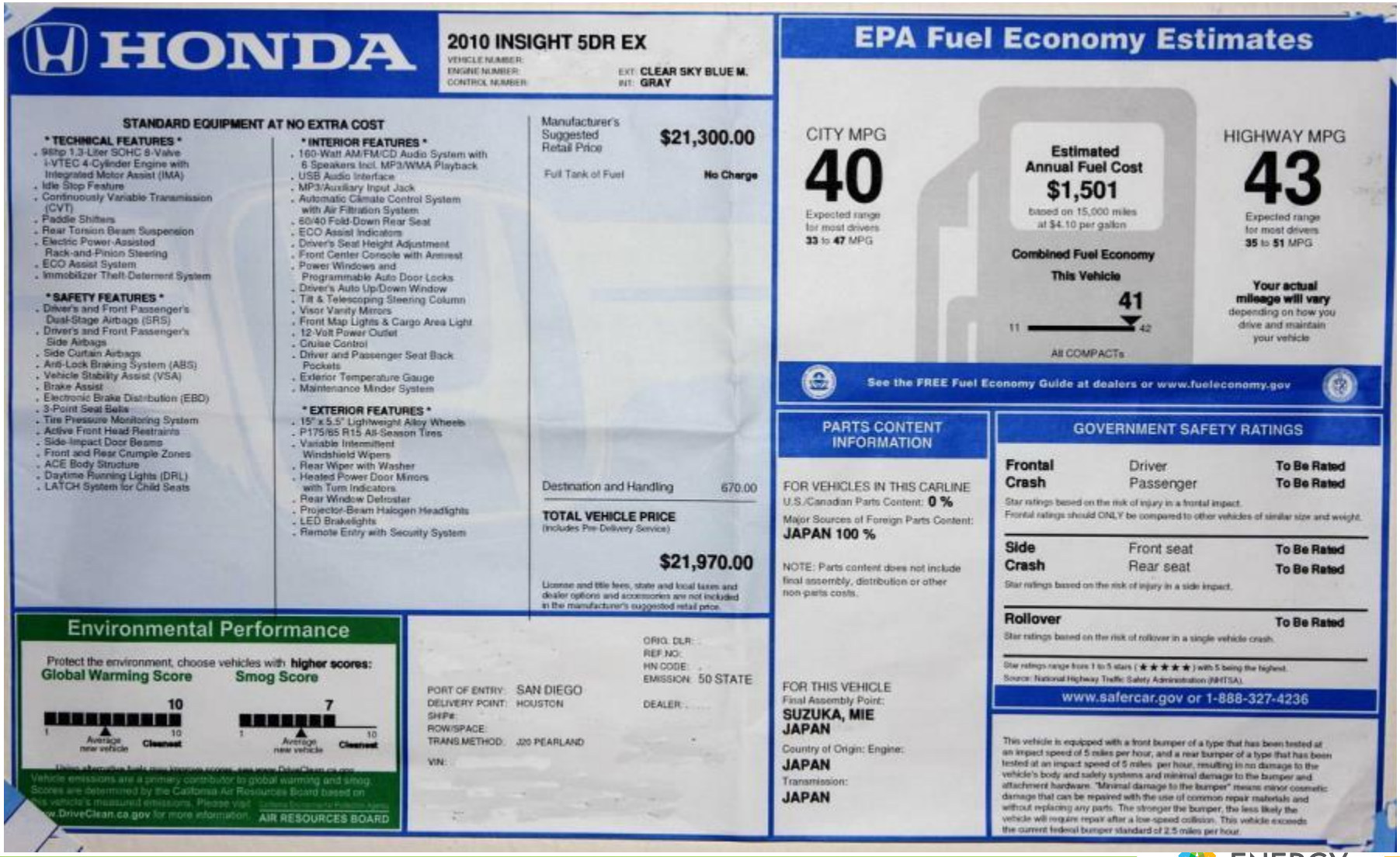


SOURCES: ROBERT H. SOCOLOW AND STEPHEN W. PACALA, PRINCETON UNIVERSITY (UPDATED REPORT); OAK RIDGE NATIONAL LABORATORY (GLOBAL CARBON EMISSIONS DATA); ICONS BY JONATHAN AVERY; GRAPHIC BY JUAN VELASCO, NGM ART

I have done it!

- Karl Kordesch 1967
- 200 miles per gal of hydrazine

Background on Vehicle Economics



HONDA

2010 INSIGHT 5DR EX

VEHICLE NUMBER:
ENGINE NUMBER:
CONTROL NUMBER:

EXT: CLEAR SKY BLUE M.
INT: GRAY

STANDARD EQUIPMENT AT NO EXTRA COST

* TECHNICAL FEATURES *
- 98hp 1.3-Liter SOHC 8-Valve i-VTEC 4-Cylinder Engine with Integrated Motor Assist (IMA)
- Idle Stop Feature
- Continuously Variable Transmission (CVT)
- Paddle Shifters
- Rear Torsion Beam Suspension
- Electric Power-Assisted Rack-and-Pinion Steering
- ECO Assist System
- Immobilizer Theft-Deterrent System

* SAFETY FEATURES *
- Driver's and Front Passenger's Dual-Stage Airbags (SRS)
- Driver's and Front Passenger's Side Airbags
- Side Curtain Airbags
- Anti-Lock Braking System (ABS)
- Vehicle Stability Assist (VSA)
- Brake Assist
- Electronic Brake Distribution (EBD)
- 3-Point Seat Belts
- Tire Pressure Monitoring System
- Active Front Head Restraints
- Side-Impact Door Beams
- Front and Rear Crumple Zones
- ACE Body Structure
- Daytime Running Lights (DRL)
- LATCH System for Child Seats

* INTERIOR FEATURES *
- 160-Watt AM/FM/CD Audio System with 6 Speakers Incl. MP3/WMA Playback
- USB Audio Interface
- MP3/Auxiliary Input Jack
- Automatic Climate Control System with Air Filtration System
- 60/40 Fold-Down Rear Seat
- ECO Assist Indicators
- Driver's Seat Height Adjustment
- Front Center Console with Armrest
- Power Windows and Programmable Auto Door Locks
- Driver's Auto Up/Down Window
- Tilt & Telescoping Steering Column
- Visor Vanity Mirrors
- Front Map Lights & Cargo Area Light
- 12-Volt Power Outlet
- Cruise Control
- Driver and Passenger Seat Back Pockets
- Exterior Temperature Gauge
- Maintenance Minder System

* EXTERIOR FEATURES *
- 15" x 5.5" Lightweight Alloy Wheels
- P175/65 R15 All-Season Tires
- Variable Intermittent Windshield Wipers
- Rear Wiper with Washer
- Heated Power Door Mirrors with Turn Indicators
- Rear Window Defroster
- Projector-Beam Halogen Headlights
- LED Brakelights
- Remote Entry with Security System

Manufacturer's Suggested Retail Price	$21,300.00
Full Tank of Fuel	No Charge
Destination and Handling	670.00
TOTAL VEHICLE PRICE (Includes Pre-Delivery Service)	$21,970.00

License and title fees, state and local taxes and dealer options and accessories are not included in the manufacturer's suggested retail price.

Environmental Performance

Protect the environment, choose vehicles with higher scores:

Global Warming Score 10 (1 – 10; Average new vehicle; Cleanest)

Smog Score 7 (1 – 10; Average new vehicle; Cleanest)

Vehicle emissions are a primary contributor to global warming and smog. Scores are determined by the California Air Resources Board based on this vehicle's measured emissions. Please visit www.DriveClean.ca.gov for more information. AIR RESOURCES BOARD

ORIG. DLR:
REF.NO:
HN CODE:
EMISSION: 50 STATE
DEALER:

PORT OF ENTRY: SAN DIEGO
DELIVERY POINT: HOUSTON
SHIP#:
ROW/SPACE:
TRANS.METHOD: J20 PEARLAND
VIN:

EPA Fuel Economy Estimates

CITY MPG 40
Expected range for most drivers 33 to 47 MPG

Estimated Annual Fuel Cost $1,501
based on 15,000 miles at $4.10 per gallon

Combined Fuel Economy
This Vehicle 41
11 – 42
All COMPACTs

HIGHWAY MPG 43
Expected range for most drivers 35 to 51 MPG

Your actual mileage will vary depending on how you drive and maintain your vehicle

See the FREE Fuel Economy Guide at dealers or www.fueleconomy.gov

PARTS CONTENT INFORMATION

FOR VEHICLES IN THIS CARLINE
U.S./Canadian Parts Content: 0 %
Major Sources of Foreign Parts Content:
JAPAN 100 %

NOTE: Parts content does not include final assembly, distribution or other non-parts costs.

FOR THIS VEHICLE
Final Assembly Point:
SUZUKA, MIE
JAPAN
Country of Origin: Engine:
JAPAN
Transmission:
JAPAN

GOVERNMENT SAFETY RATINGS

Frontal Crash	Driver	To Be Rated
	Passenger	To Be Rated

Star ratings based on the risk of injury in a frontal impact.
Frontal ratings should ONLY be compared to other vehicles of similar size and weight.

Side Crash	Front seat	To Be Rated
	Rear seat	To Be Rated

Star ratings based on the risk of injury in a side impact.

Rollover To Be Rated
Star ratings based on the risk of rollover in a single vehicle crash.

Star ratings range from 1 to 5 stars (★★★★★) with 5 being the highest.
Source: National Highway Traffic Safety Administration (NHTSA).

www.safercar.gov or 1-888-327-4236

This vehicle is equipped with a front bumper of a type that has been tested at an impact speed of 5 miles per hour, and a rear bumper of a type that has been tested at an impact speed of 5 miles per hour, resulting in no damage to the vehicle's body and safety systems and minimal damage to the bumper and attachment hardware. "Minimal damage to the bumper" means minor cosmetic damage that can be repaired with the use of common repair materials and without replacing any parts. The stronger the bumper, the less likely the vehicle will require repair after a low-speed collision. This vehicle exceeds the current federal bumper standard of 2.5 miles per hour.

What are the costs of vehicle ownership?

Interaction Required

- Purchase price (ev vs cv)
- Fuel costs (ev vs cv)
- Maintenance (ev vs cv)
- Insurance (ev vs cv)
- Tires and breakpads (maintenance) (ev vs cv)
- Unforseen repair (ev vs cv)
- Registration (ev vs cv)
- Tolls (cv vs ev)
- Taxes (ev vs cv)
- Road use fees
- Parking (ev vs cv)

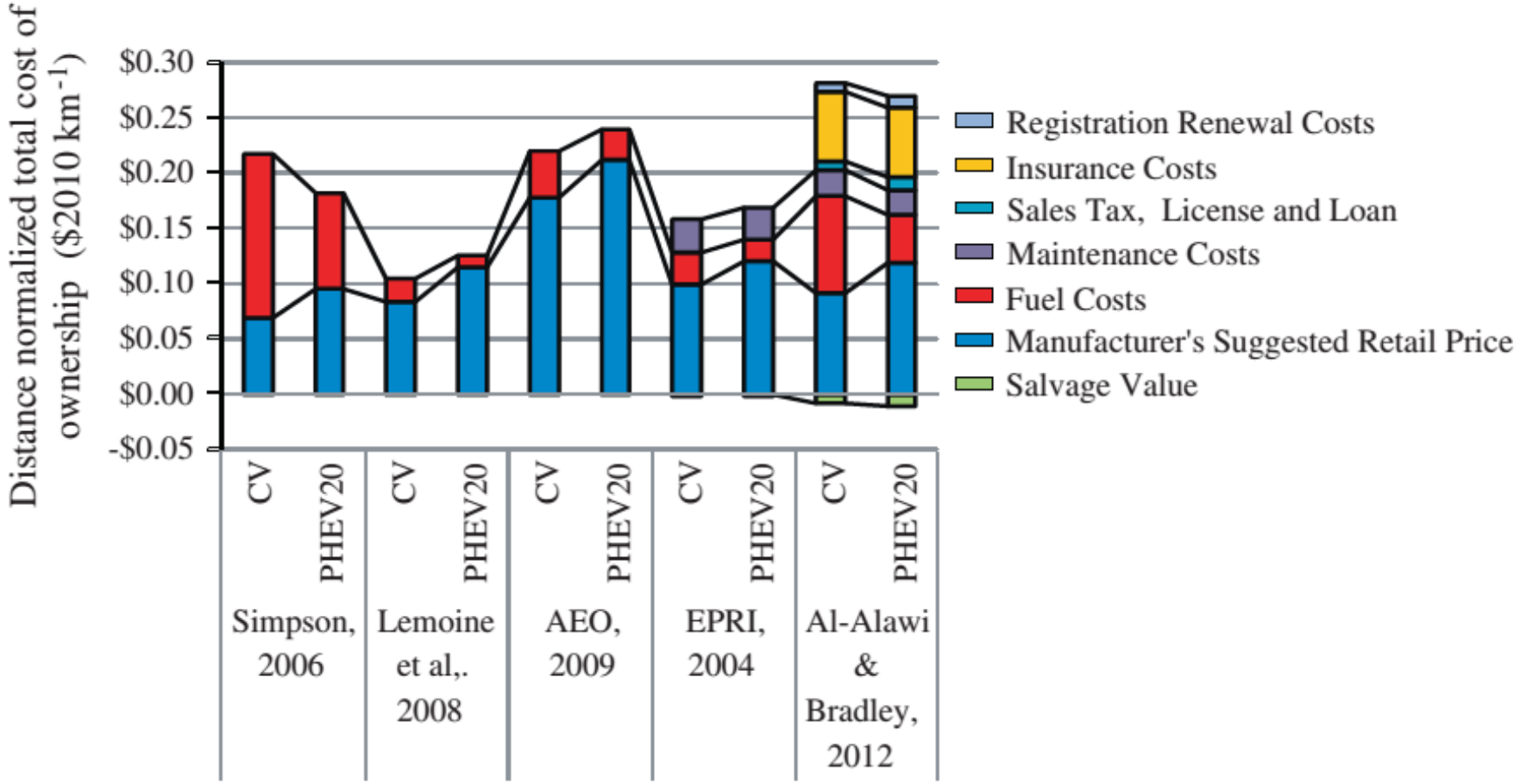
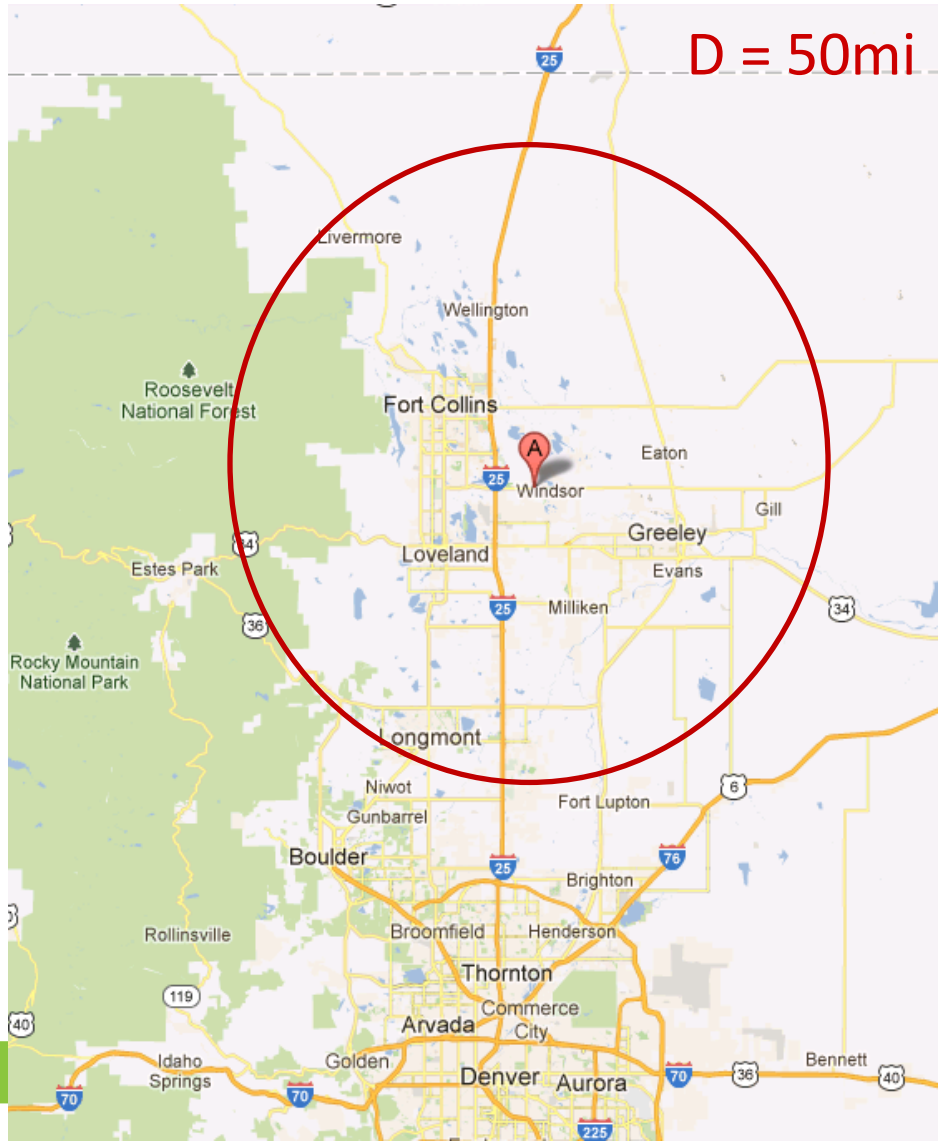


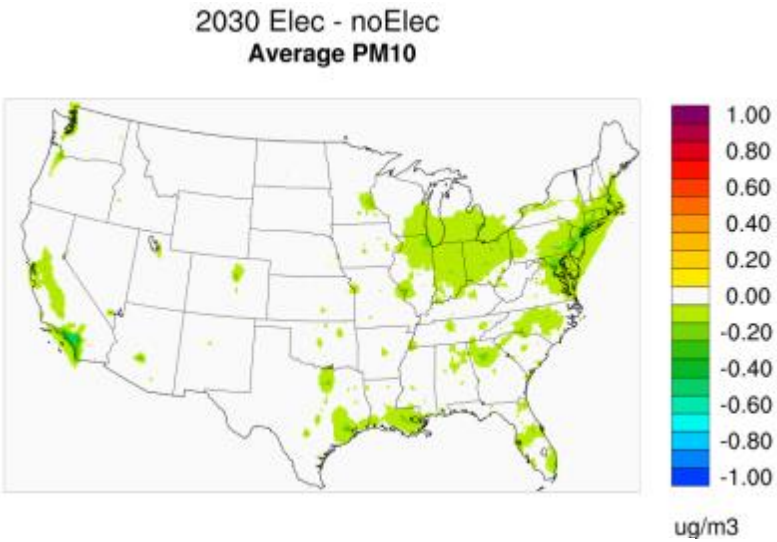
Fig. 1. Total cost in $2010 per kilometer using each study's parameters and TCO model.

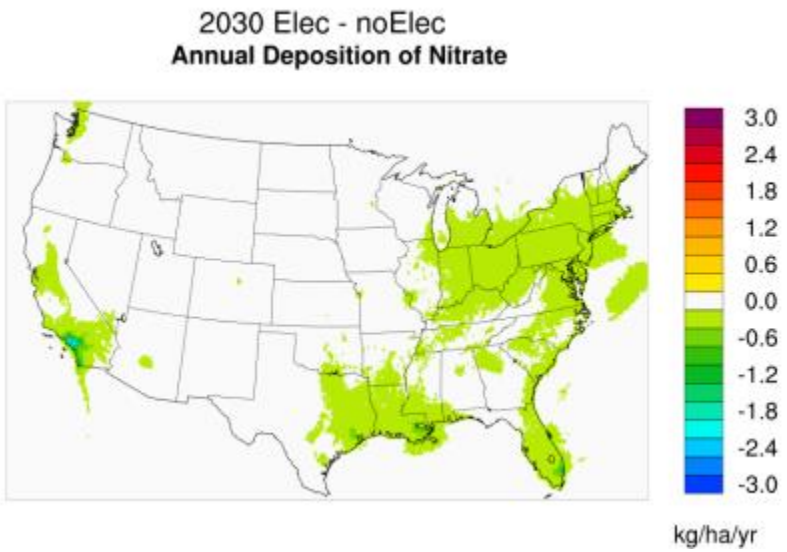
Plug in Electric Vehicles

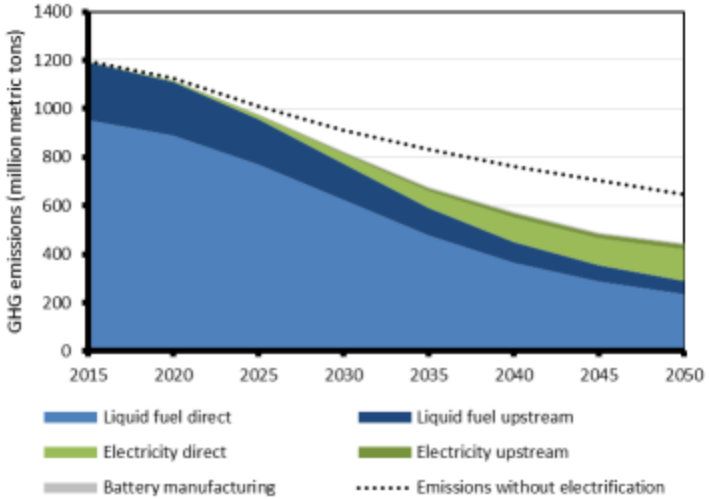
- Plug in Hybrid Vehicles
  - Chevrolet Volt
    - 40 mi of EV range
  - Toyota Prius Plug in
    - 12 mi of EV range
- Charge at home
- Fuel with Gasoline for long trips



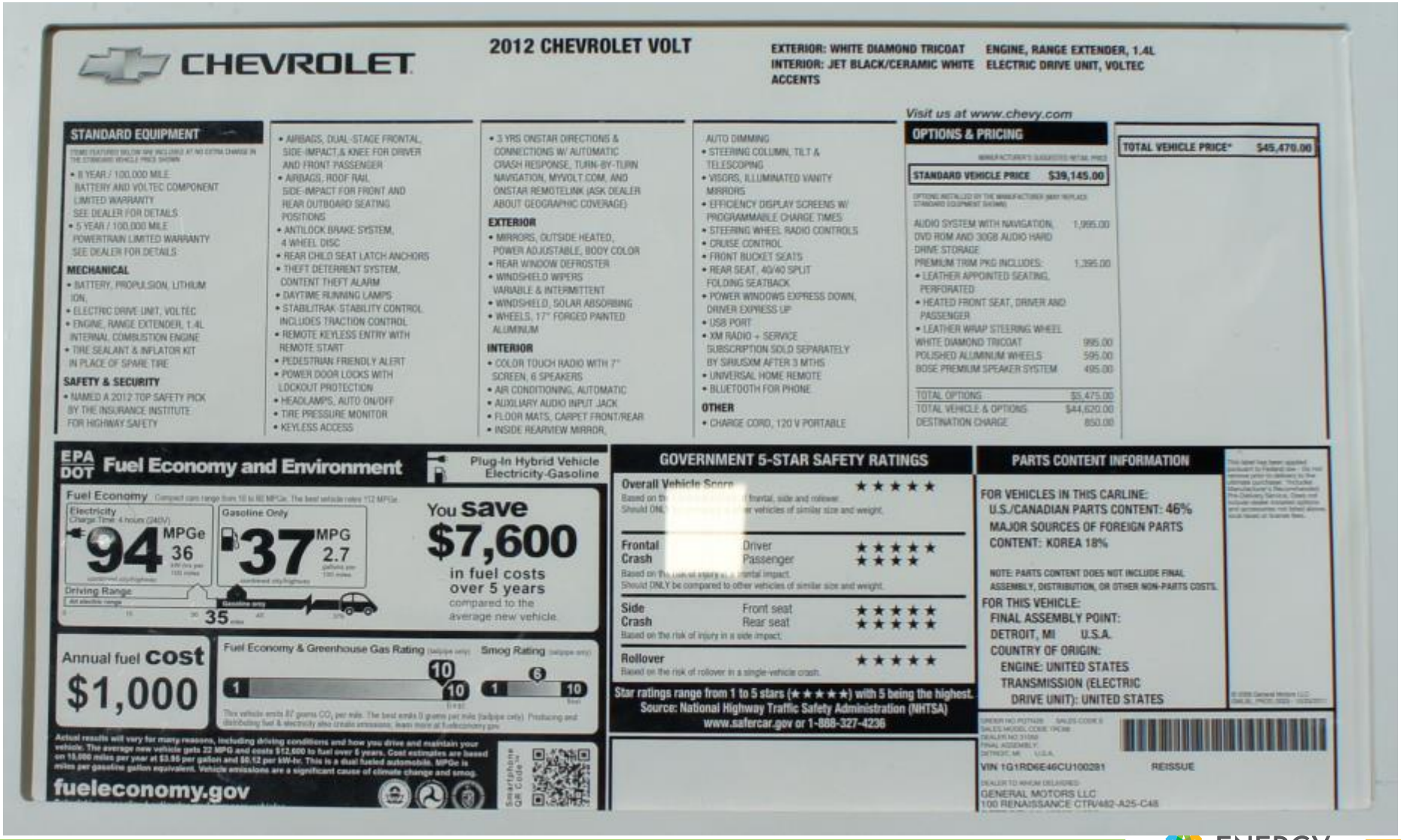
PHEVs are good







What are the costs of PHEV ownership?



CHEVROLET

2012 CHEVROLET VOLT

EXTERIOR: WHITE DIAMOND TRICOAT
INTERIOR: JET BLACK/CERAMIC WHITE ACCENTS

ENGINE, RANGE EXTENDER, 1.4L
ELECTRIC DRIVE UNIT, VOLTEC

Visit us at www.chevy.com

STANDARD EQUIPMENT

- 8 YEAR / 100,000 MILE BATTERY AND VOLTEC COMPONENT LIMITED WARRANTY SEE DEALER FOR DETAILS
- 5 YEAR / 100,000 MILE POWERTRAIN LIMITED WARRANTY SEE DEALER FOR DETAILS

MECHANICAL

- BATTERY, PROPULSION, LITHIUM ION,
- ELECTRIC DRIVE UNIT, VOLTEC
- ENGINE, RANGE EXTENDER, 1.4L INTERNAL COMBUSTION ENGINE
- TIRE SEALANT & INFLATOR KIT IN PLACE OF SPARE TIRE

SAFETY & SECURITY

- NAMED A 2012 TOP SAFETY PICK BY THE INSURANCE INSTITUTE FOR HIGHWAY SAFETY
- AIRBAGS, DUAL-STAGE FRONTAL, SIDE-IMPACT & KNEE FOR DRIVER AND FRONT PASSENGER
- AIRBAGS, ROOF RAIL SIDE-IMPACT FOR FRONT AND REAR OUTBOARD SEATING POSITIONS
- ANTILOCK BRAKE SYSTEM, 4 WHEEL DISC
- REAR CHILD SEAT LATCH ANCHORS
- THEFT DETERRENT SYSTEM, CONTENT THEFT ALARM
- DAYTIME RUNNING LAMPS
- STABILITRAK STABILITY CONTROL INCLUDES TRACTION CONTROL
- REMOTE KEYLESS ENTRY WITH REMOTE START
- PEDESTRIAN FRIENDLY ALERT
- POWER DOOR LOCKS WITH LOCKOUT PROTECTION
- HEADLAMPS, AUTO ON/OFF
- TIRE PRESSURE MONITOR
- KEYLESS ACCESS
- 3 YRS ONSTAR DIRECTIONS & CONNECTIONS W/ AUTOMATIC CRASH RESPONSE, TURN-BY-TURN NAVIGATION, MYVOLT.COM, AND ONSTAR REMOTELINK (ASK DEALER ABOUT GEOGRAPHIC COVERAGE)

EXTERIOR

- MIRRORS, OUTSIDE HEATED, POWER ADJUSTABLE, BODY COLOR
- REAR WINDOW DEFROSTER
- WINDSHIELD WIPERS VARIABLE & INTERMITTENT
- WINDSHIELD, SOLAR ABSORBING
- WHEELS, 17" FORGED PAINTED ALUMINUM

INTERIOR

- COLOR TOUCH RADIO WITH 7" SCREEN, 6 SPEAKERS
- AIR CONDITIONING, AUTOMATIC
- AUXILIARY AUDIO INPUT JACK
- FLOOR MATS, CARPET FRONT/REAR
- INSIDE REARVIEW MIRROR, AUTO DIMMING
- STEERING COLUMN, TILT & TELESCOPING
- VISORS, ILLUMINATED VANITY MIRRORS
- EFFICIENCY DISPLAY SCREENS W/ PROGRAMMABLE CHARGE TIMES
- STEERING WHEEL RADIO CONTROLS
- CRUISE CONTROL
- FRONT BUCKET SEATS
- REAR SEAT, 40/40 SPLIT FOLDING SEATBACK
- POWER WINDOWS EXPRESS DOWN, DRIVER EXPRESS UP
- USB PORT
- XM RADIO + SERVICE SUBSCRIPTION SOLD SEPARATELY BY SIRIUSXM AFTER 3 MTHS
- UNIVERSAL HOME REMOTE
- BLUETOOTH FOR PHONE

OTHER

- CHARGE CORD, 120 V PORTABLE

OPTIONS & PRICING

STANDARD VEHICLE PRICE	$39,145.00
AUDIO SYSTEM WITH NAVIGATION, DVD ROM AND 30GB AUDIO HARD DRIVE STORAGE	1,995.00
PREMIUM TRIM PKG INCLUDES: • LEATHER APPOINTED SEATING, PERFORATED • HEATED FRONT SEAT, DRIVER AND PASSENGER • LEATHER WRAP STEERING WHEEL	1,395.00
WHITE DIAMOND TRICOAT	995.00
POLISHED ALUMINUM WHEELS	595.00
BOSE PREMIUM SPEAKER SYSTEM	495.00
TOTAL OPTIONS	$5,475.00
TOTAL VEHICLE & OPTIONS	$44,620.00
DESTINATION CHARGE	850.00
TOTAL VEHICLE PRICE*	$45,470.00

EPA DOT Fuel Economy and Environment

Plug-In Hybrid Vehicle Electricity-Gasoline

Fuel Economy

Electricity Charge Time: 4 hours (240V) 94 MPGe 36 kW-hrs per 100 miles

Gasoline Only 37 MPG 2.7 gallons per 100 miles

Driving Range 35 miles

You save $7,600 in fuel costs over 5 years compared to the average new vehicle.

Annual fuel cost $1,000

Fuel Economy & Greenhouse Gas Rating (vehicle only) 10 / 10

Smog Rating (vehicle only) 6 / 10

This vehicle emits 87 grams CO_2 per mile. The best emits 0 grams per mile (vehicle only). Producing and distributing fuel & electricity also create emissions; learn more at fueleconomy.gov

Actual results will vary for many reasons, including driving conditions and how you drive and maintain your vehicle. The average new vehicle gets 22 MPG and costs $12,600 to fuel over 5 years. Cost estimates are based on 15,000 miles per year at $3.95 per gallon and $0.12 per kW-hr. This is a dual fueled automobile. MPGe is miles per gasoline gallon equivalent. Vehicle emissions are a significant cause of climate change and smog.

fueleconomy.gov

Smartphone QR Code™

GOVERNMENT 5-STAR SAFETY RATINGS

Overall Vehicle Score ★★★★★
Based on the [illegible] frontal, side and rollover
Should ONLY be compared to other vehicles of similar size and weight.

Frontal Crash — Driver ★★★★★ — Passenger ★★★★
Based on the [illegible] frontal impact.
Should ONLY be compared to other vehicles of similar size and weight.

Side Crash — Front seat ★★★★★ — Rear seat ★★★★★
Based on the risk of injury in a side impact.

Rollover ★★★★★
Based on the risk of rollover in a single-vehicle crash.

Star ratings range from 1 to 5 stars (★★★★★) with 5 being the highest.
Source: National Highway Traffic Safety Administration (NHTSA)
www.safercar.gov or 1-888-327-4236

PARTS CONTENT INFORMATION

FOR VEHICLES IN THIS CARLINE:
U.S./CANADIAN PARTS CONTENT: 46%
MAJOR SOURCES OF FOREIGN PARTS CONTENT: KOREA 18%

NOTE: PARTS CONTENT DOES NOT INCLUDE FINAL ASSEMBLY, DISTRIBUTION, OR OTHER NON-PARTS COSTS.

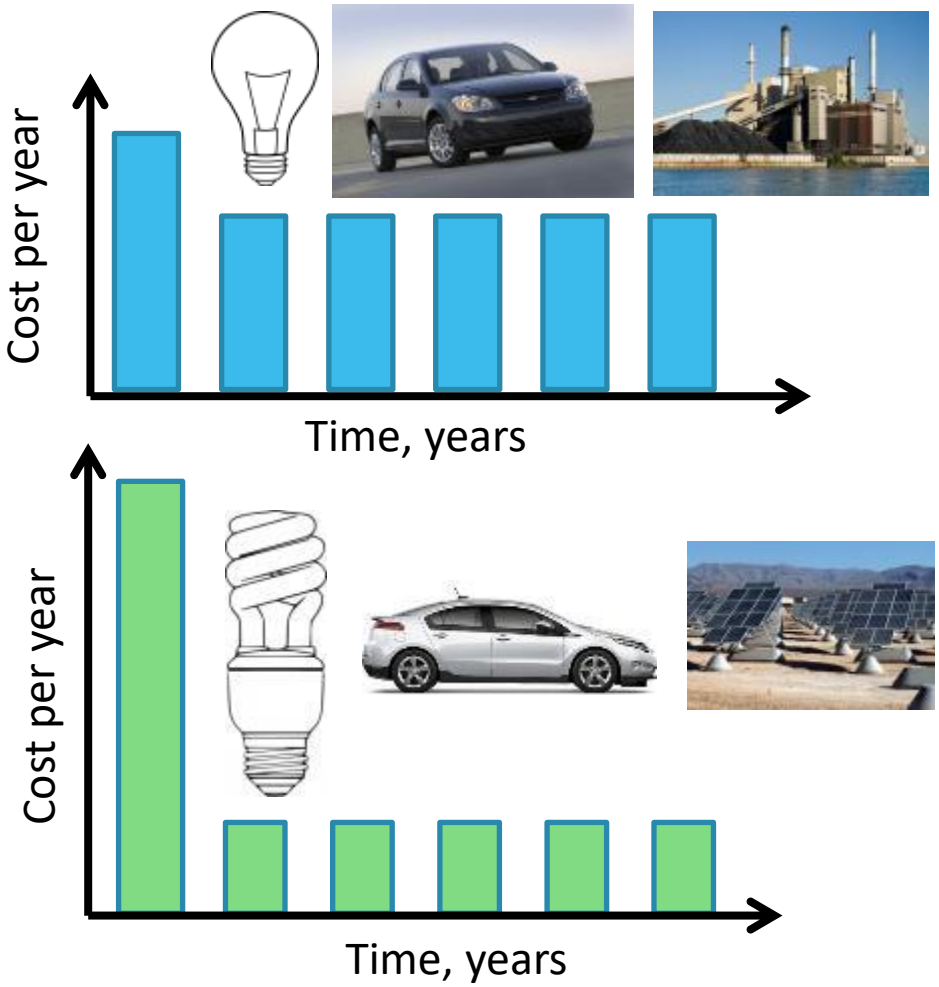
FOR THIS VEHICLE:
FINAL ASSEMBLY POINT:
DETROIT, MI U.S.A.
COUNTRY OF ORIGIN:
ENGINE: UNITED STATES
TRANSMISSION (ELECTRIC DRIVE UNIT): UNITED STATES

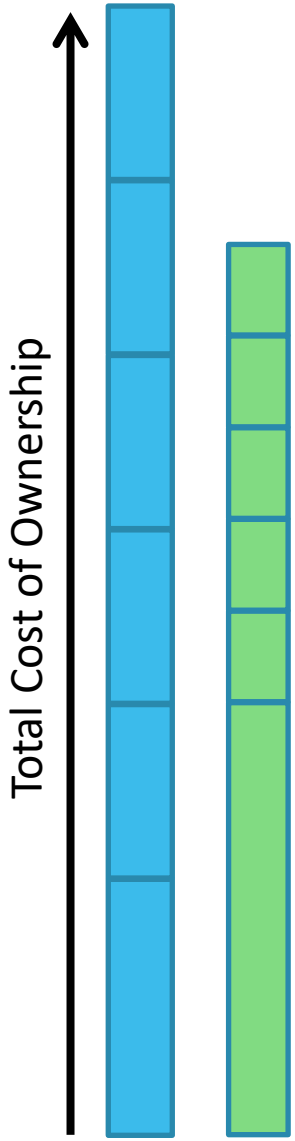
VIN 1G1RD6E46CU100281 REISSUE

GENERAL MOTORS LLC
100 RENAISSANCE CTR/482-A25-C48

Economics of "better technologies"

- Purchase price vs. operating cost





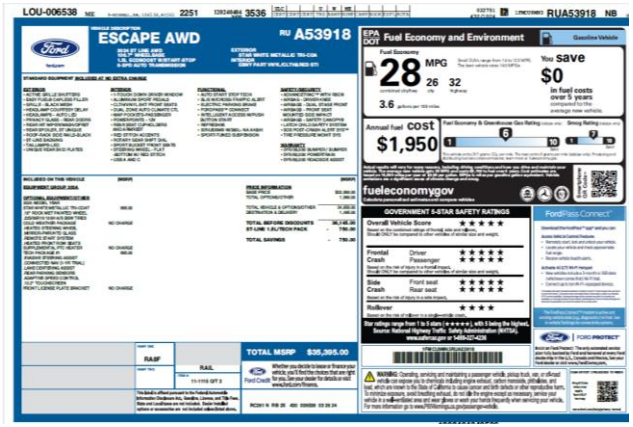
Economics of "better technologies"

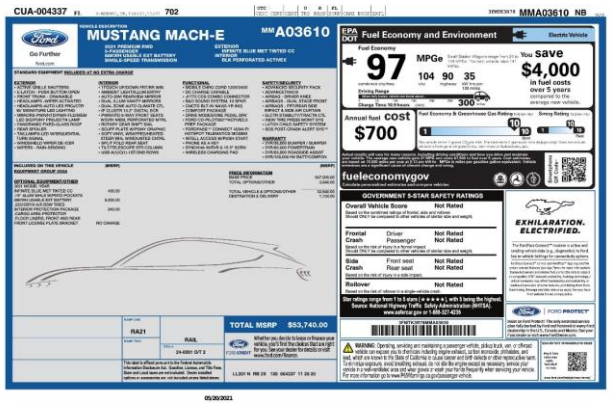
$$DCF = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \cdots + \frac{CF_n}{(1+r)^n}$$

$$FV = DCF \cdot (1+r)^n$$

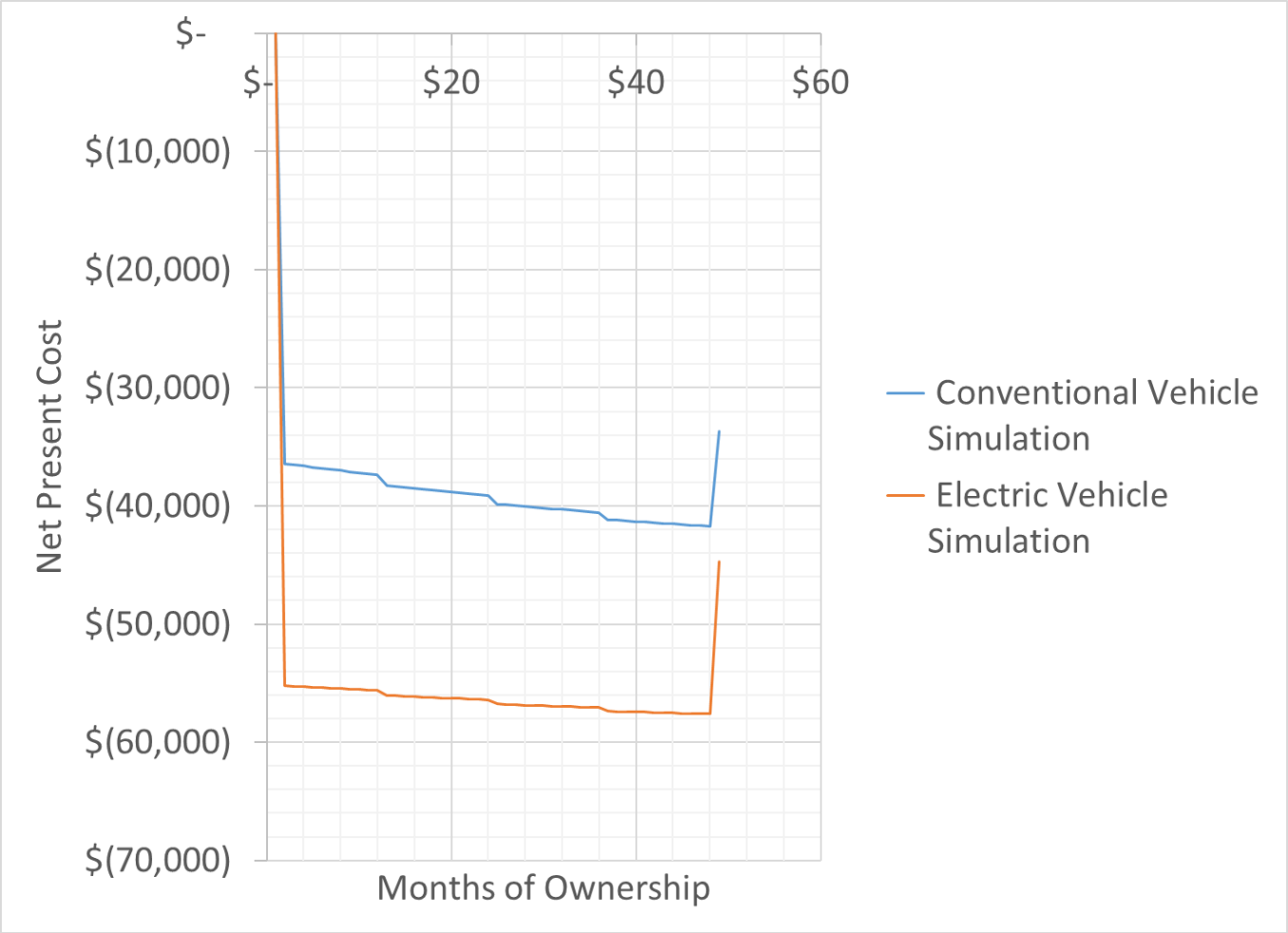
- Cash Flow1 is the difference in cost of ownership between the EV and the CV in Year 1
- What is r?
  - A 6% discount rate represents a 5% loan in constant dollars

Not in Stock – In Transit


LOU-006538
RUA53918
ESCAPE AWD
A53918
Fuel Economy and Environment
Gasoline Vehicle
28 MPG
26
32
3.6
You save $0 in fuel costs over 5 years compared to the average new vehicle.
Annual fuel cost $1,950
fueleconomygov
GOVERNMENT 5-STAR SAFETY RATINGS
Overall Vehicle Score
Frontal Crash
Driver
Passenger
Side Crash
Front seat
Rear seat
Rollover
TOTAL MSRP $35,395.00

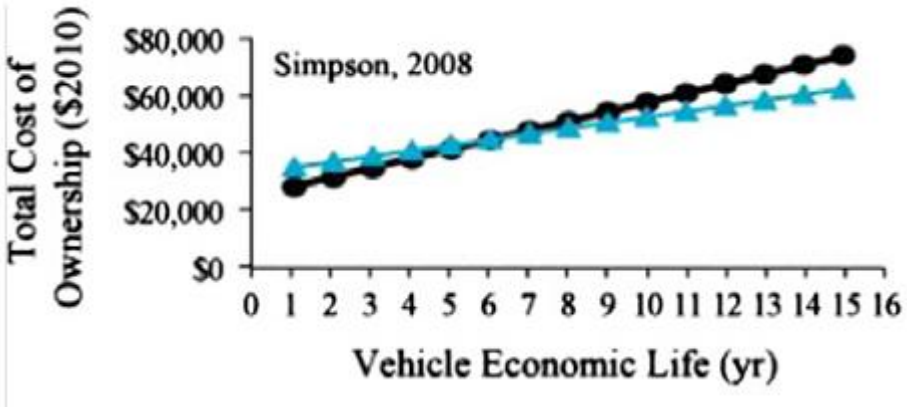

CUA-004337
MMA03610
MUSTANG MACH-E
A03610
Fuel Economy and Environment
Electric Vehicle
97 MPGe
104
90
35
You save $4,000 in fuel costs over 5 years compared to the average new vehicle.
Annual fuel cost $700
fueleconomygov
GOVERNMENT 5-STAR SAFETY RATINGS
Overall Vehicle Score Not Rated
Frontal Crash
Driver Not Rated
Passenger Not Rated
Side Crash
Front seat Not Rated
Rear seat Not Rated
Rollover Not Rated
EXHILARATION. ELECTRIFIED.
TOTAL MSRP $53,740.00
05/20/2021

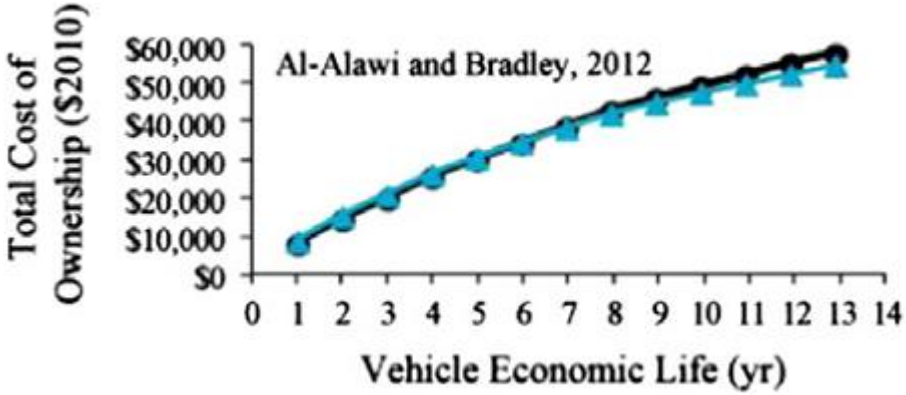
NPV calcs for that Ford Mach E Comparison

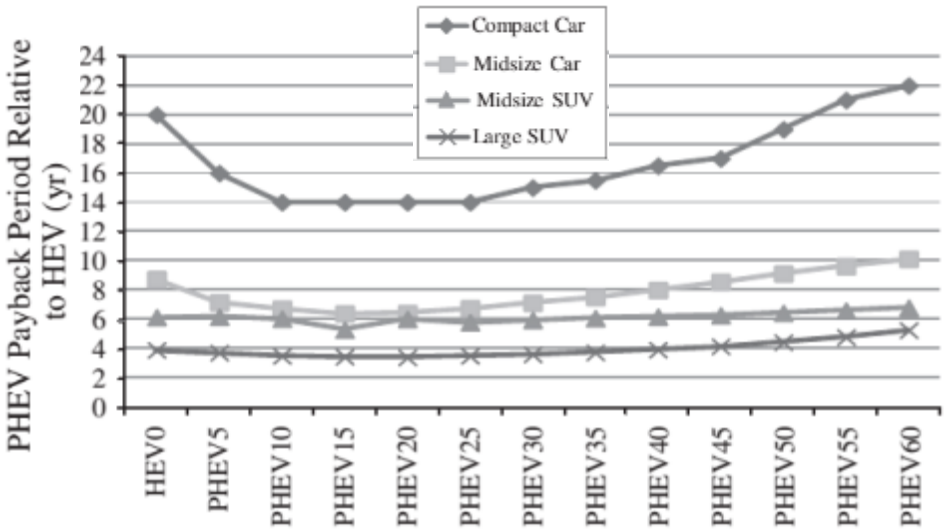


Metrics of Decision Making

- "Payback period" is the most common metric of binary decision making for engineering and energy efficiency







Metrics of Decision Making

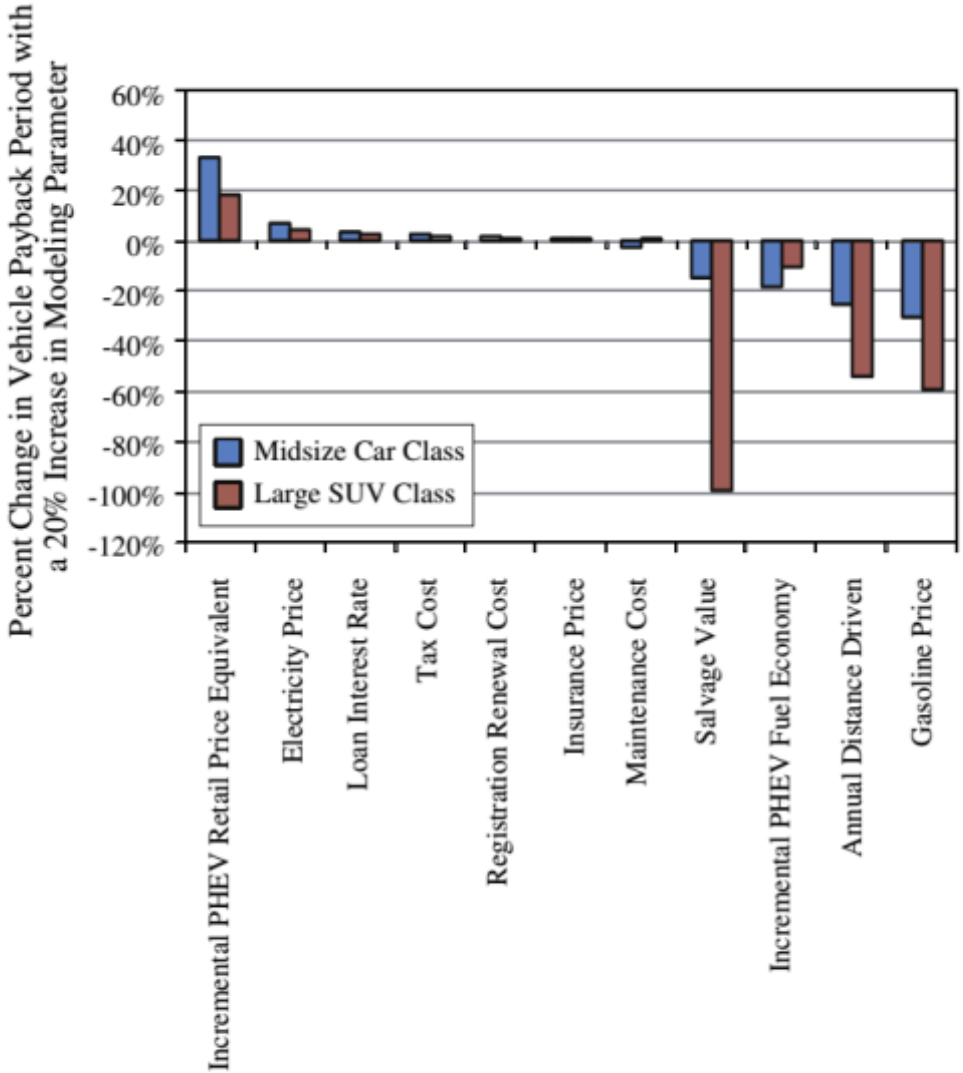
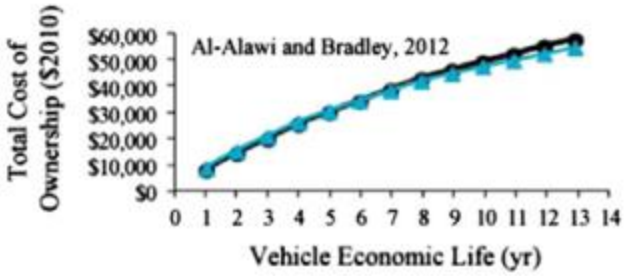


Fig. 5. Sensitivity of the PHEV20 economic payback period to TCO model parameters in the Mid-Sized Car and large SUV classes compared to CV.

- Payback period is a very sensitive metric of economic performance
- What might be better metrics ?



Thank You

Contact

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ENERGY INSTITUTE
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